

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12396773
Kind Code	B1
Date of Patent	August 26, 2025
Inventor(s)	Wu; Shuce et al.

Automatic bone cement mixing and injection device

Abstract

An automatic bone cement mixing and injection device, comprising a cylindrical structure, a sliding block, a mixer, pistons, a mixer driving device and a piston driving device; two ends of the cylindrical structure are respectively an open installation port and a narrowed injection port, a feed port is arranged on one side wall of the cylindrical structure; one end of the mixer passes through the installation port of the cylindrical structure and is placed inside the cylindrical structure, and the other end is connected to a rotating shaft of the mixer driving device; one end of each piston passes through the installation port of the cylindrical structure and is placed inside the cylindrical structure, and the other end of each piston is pressed against the sliding block; the sliding block is connected to the piston driving device and is driven by the piston driving device to make linear motion.

Inventors:	Wu; Shuce (Jiangsu, CN), Cui; Qinglei (Jiangsu, CN)
Applicant:	Suzhou Feima Medical Technology Co., Ltd. (Jiangsu, CN)
Family ID:	1000008490649
Appl. No.:	19/048793
Filed:	February 07, 2025

Foreign Application Priority Data

CN	202410200317.3	Feb. 23, 2024
----	----------------	---------------

Publication Classification

Int. Cl.: A61B17/88 (20060101)

U.S. Cl.:

CPC A61B17/8833 (20130101); A61B2017/8838 (20130101)

Field of Classification Search

CPC: A61B (17/8833); A61B (2017/8838)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4277184	12/1980	Solomon	366/139	A61B 17/8825
5842786	12/1997	Solomon	366/139	B01F 31/40
8167835	12/2011	Keller	604/82	B05C 17/01
11737802	12/2022	Williams	606/94	A61B 17/8827
2019/0038331	12/2018	Purdy et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
203369946	12/2013	CN	N/A
105147384	12/2014	CN	N/A
109069753	12/2017	CN	N/A
209392078	12/2018	CN	N/A
212699095	12/2020	CN	N/A
212819279	12/2020	CN	N/A
215463513	12/2021	CN	N/A
219048898	12/2022	CN	N/A
219354127	12/2022	CN	N/A
219645855	12/2022	CN	N/A
117426852	12/2023	CN	N/A
20210037869	12/2020	KR	N/A

Primary Examiner: Woodall; Nicholas W

Background/Summary

1. TECHNICAL FIELD

(1) The invention relates to the technical field of medical surgery, in particular to an automatic bone cement mixing and injection device.

2. BACKGROUND ART

(2) Bone cement, as a widely used medical material in orthopedic surgery, is named for its unique physical properties and its appearance and shape after setting, which resemble those of ordinary white cement used in construction. As a common implant in the body, bone cement has a wide range of applications in the medical field, playing a key role particularly in procedures such as vertebroplasty and joint replacement surgery. With the continuous improvement of medical levels in China, the usage of bone cement has also shown a trend of year-on-year growth.

(3) The use of bone cement primarily involves two crucial processes: mixing and injection. Prior to the surgery, the doctor needs to combine the solid and liquid ingredients and manually mix them for several minutes to ensure uniform mixing of the bone cement. Following this, the doctor transfers the well-mixed bone cement to a syringe, preparing it for injection.

(4) However, the existing bone cement mixing devices currently developed by researchers still have some shortcomings. Firstly, most of these devices rely on manual operation, using tools with mixing blades, which is both time-consuming and laborious. Secondly, these devices typically only have mixing functionality and lack injection capabilities, meaning that doctors need to switch tools after mixing to perform the injection, adding complexity to the operation. Finally, since the injection of bone cement requires precise control of the injection volume, doctors often need to frequently move between the operating table and the medical imaging room to control the injection volume by observing the images; this approach not only prolongs the surgery time, but also brings inconvenience to the medical staff's operation, increasing the exposure of operators and/or patients to radiation.

(5) In order to address these problems, patents CN202221323452.X and CN201120395926.7 propose two bone cement mixing devices. Patent CN202221323452.X presents a micro electric mixer, which uses a motor to drive spiral blades to rotate, transporting the raw materials from the bottom of the centrifuge tube to the upper end for mixing, thus improving mixing efficiency. However, this device still has some limitations, such as the inability to adjust the mixing speed, its unsuitability for direct injection operations, and the lack of remote operation capabilities.

(6) On the other hand, patent CN201120395926.7 introduces a convenient mixing and injection device for bone cement. This device cleverly combines the mixing frame and the syringe body, achieving an integrated operation for both mixing and injection. The bone cement can be mixed by rotating the mixing handle, while pressing the handle downward allows for injection. This design simplifies the operation process and improves surgical efficiency. However, the device also has some drawbacks, such as both mixing and injection operations being manual, the inability to achieve automation, and the lack of remote operation capabilities.

(7) In summary, although these two bone cement mixing devices have solved the problems existing in the traditional surgical preparation process to a certain extent, there are still some limitations that need to be further improved and perfected.

3. SUMMARY OF THE INVENTION

(8) The object of the invention is to propose an automatic bone cement mixing and injection device, which features automatic adjustment of mixing speed and integrates both mixing and injection functions, and can meet a wider range of clinical needs, improve surgical efficiency, and enhance the convenience and effectiveness of bone cement surgeries.

(9) In order to realize the above objects, the technical scheme of the invention is as follows: an automatic bone cement mixing and injection device, comprising a cylindrical structure, a sliding block, a mixer, pistons, a mixer driving device and a piston driving device; two ends of the cylindrical structure are respectively an open installation port and a narrowed injection port, a feed port is arranged on one side wall of the cylindrical structure; one end of the mixer passes through the installation port of the cylindrical structure and is placed inside the cylindrical structure, and the other end is connected to a rotating shaft of the mixer driving device; one end of each piston passes through the installation port of the cylindrical structure and is placed inside the cylindrical structure, and the other end of each piston is pressed against the sliding block; the sliding block is connected to the piston driving device and is driven by the piston driving device to make linear motion.

(10) The sliding block is composed of an upper sliding block, a lower sliding block, a bearing and a retaining ring; the upper sliding block and the lower sliding block are provided with grooves that match an outer ring of the bearing; the bearing is placed in the grooves, and the retaining ring is pressed against the bearing; the other end of each piston is pressed against the retaining ring of the sliding block.

(11) The installation port is provided with an end cover that cooperates with the mixer and the pistons and seals the installation port.

(12) A plurality of blades are arranged at one end of the mixer installed inside the cylindrical

structure; the blades are arranged along an axial direction of the mixer; an installation and movement space for each piston is formed between two adjacent blades and a cylinder wall of the cylindrical structure.

(13) The blades are provided with through holes to facilitate the mixing of bone cement.

(14) One end of each piston installed inside the cylindrical structure is provided with a columnar structure that fits tightly with the adjacent blades and the cylinder wall.

(15) A nut is fixed on the sliding block; a screw rod is fixed on an output shaft of the piston driving device; the nut is installed on the screw rod. In this process, the cooperation between the nut and the sliding block ensures that each piston only performs linear motion.

(16) The working principle of the invention is as follows: mixing action: when the mixer driving device is activated, it drives the mixer and the pistons mounted on it to rotate, thereby achieving uniform mixing of the bone cement; injection action: when the piston driving device is activated, a series of mechanical connections transfer linear motion to the three pistons, pushing the bone cement along the axial direction of the mixer for injection.

(17) The invention integrates the mixing and injection of bone cement through a uniquely designed combination of the pistons and mixer; it features a high degree of automation, using electric drive for both the mixing and injection processes, thereby reducing the overall surgery time, lightening the physical burden on the operator, and minimizing the exposure of both the operator and/or the patient to radiation.

Description

4. BRIEF DESCRIPTION OF ACCOMPANYING DRAWINGS

(1) FIG. 1 is a schematic diagram of the assembly of the components of the invention.

(2) FIG. 2 is a cross-sectional view of the A-A plane in FIG. 1.

(3) FIG. 3 is an exploded view of the pistons and mixer in the invention.

5. SPECIFIC EMBODIMENT OF THE INVENTION

(4) In order to make the objects, technical schemes and advantages of the embodiments of the invention clearer, the technical schemes in the embodiments of the invention will be clearly and completely described below in combination with the accompanying drawings in the embodiments of the invention, obviously, the described embodiments are some, but not all embodiments of the invention. The components of the embodiments of the invention generally described and illustrated in the drawings herein may be arranged and designed in a variety of different configurations. Therefore, the following detailed description of the embodiments of the invention provided in the accompanying drawings is not intended to limit the scope of the claimed invention, but is merely representative of selected embodiments of the invention. Based on the embodiments in this invention, all other embodiments obtained by those of ordinary skill in the art without making creative efforts shall fall within the protection scope of this invention.

(5) As shown in FIGS. 1 and 2, an automatic bone cement mixing and injection device of the invention comprises a cylindrical structure **100**, a sliding block **200**, a mixer **300**, pistons **400**, a mixer driving device **500** and a piston driving device **600**; two ends of the cylindrical structure **100** are respectively an open installation port **110** and a narrowed injection port **120**, a feed port **130** is arranged on one side wall of the cylindrical structure **100**, and the feed port **130** can be completely sealed after the feeding is completed; one end of the mixer **300** passes through the installation port **110** of the cylindrical structure **100** and is placed inside the cylindrical structure **100**, and the other end is connected to a rotating shaft of the mixer driving device **500**, this connection can be direct or indirect, this connection method in this embodiment is to connect through a coupling **700**; the sliding block **200** is connected to the piston driving device **600** and is driven by the piston driving device **600** to make linear motion; the sliding block **200** is composed of an upper sliding block **210**,

a lower sliding block **220**, a bearing **230** and a retaining ring **240**; the upper sliding block **210** and the lower sliding block **220** are provided with grooves that match an outer ring of the bearing **230**; the bearing **230** is placed in the grooves, and the retaining ring **240** is pressed against the bearing **230**; one end of each piston **400** passes through the installation port **110** of the cylindrical structure **100** and is placed inside the cylindrical structure **100**, and the other end of each piston **400** is pressed against the retaining ring **240** of the sliding block **200**.

(6) The installation port **110** is provided with an end cover that cooperates with the mixer **300** and the pistons **400** and seals the installation port **110**; the end cover in this embodiment is composed of a left end cover **1101** and a right end cover **1102**.

(7) A plurality of blades **310** are arranged at one end of the mixer **300** installed inside the cylindrical structure **100**; in this embodiment, three blades **310** are provided; the blades **310** are arranged along an axial direction of the mixer **300**; an installation and movement space for each piston **400** is formed between two adjacent blades **310** and a cylinder wall of the cylindrical structure **100**.

(8) The blades **310** are provided with through holes **3101** to facilitate the mixing of bone cement.

(9) One end of each piston **400** installed inside the cylindrical structure **100** is provided with a columnar structure **410** that fits tightly with the adjacent blades **310** and the cylinder wall.

(10) A nut **250** is fixed on the sliding block **200**, and the nut **250** in this embodiment is fixed below the lower sliding block **220**; a screw rod **610** is fixed on an output shaft of the piston driving device **600**, and the nut **250** is installed on the screw rod **610**. In this process, the cooperation between the nut **250** and the sliding block **200** ensures that each piston **400** only performs linear motion.

(11) The mixer driving device **500** and the piston driving device **600** in this embodiment both use electric motors, but this choice does not limit the protection scope of the invention. Any device that can achieve the same technical effect and manual operation are within the protection scope claimed by the invention.

(12) In another embodiment of the invention, a remote wireless operating device may be included. The control device arranged on the main body of the automatic bone cement mixing and injection device of the invention receives the signal of the remote wireless operating device and controls the mixer driving device and the piston driving device according to the instruction. For those skilled in the art, the remote wireless operating device and the control device are based on existing electrical and automation technologies.

(13) The above description is only a preferred embodiment of the invention and is not intended to limit the invention. For those skilled in the art, the invention may have various modifications and variations. Any modification, equivalent substitution, improvement, etc. made within the spirit and principle of the invention shall be included in the protection scope of the invention.

Claims

1. An automatic bone cement mixing and injection device, comprising a cylindrical structure, a sliding block, a mixer, pistons, a mixer driving device and a piston driving device; two ends of the cylindrical structure are respectively an open installation port and a narrowed injection port, a feed port is arranged on one side wall of the cylindrical structure; one end of the mixer passes through the installation port of the cylindrical structure and is placed inside the cylindrical structure, and the other end is connected to a rotating shaft of the mixer driving device; one end of each piston passes through the installation port of the cylindrical structure and is placed inside the cylindrical structure, and the other end of each piston is pressed against the sliding block; the sliding block is connected to the piston driving device and is driven by the piston driving device to make linear motion; a plurality of blades are arranged at one end of the mixer installed inside the cylindrical structure; the blades are arranged along an axial direction of the mixer; an installation and movement space for each piston is formed between two adjacent blades and a cylinder wall of the

cylindrical structure.

2. The automatic bone cement mixing and injection device of claim 1, wherein the sliding block is composed of an upper sliding block, a lower sliding block, a bearing and a retaining ring; the upper sliding block and the lower sliding block are provided with grooves that match an outer ring of the bearing; the bearing is placed in the grooves, and the retaining ring is pressed against the bearing; the other end of each piston is pressed against the retaining ring of the sliding block.

3. The automatic bone cement mixing and injection device of claim 1, wherein the installation port is provided with an end cover that cooperates with the mixer and the pistons and seals the installation port.

4. The automatic bone cement mixing and injection device of claim 1, wherein the blades are provided with through holes to facilitate the mixing of bone cement.

5. The automatic bone cement mixing and injection device of claim 1, wherein one end of each piston installed inside the cylindrical structure is provided with a columnar structure that fits tightly with the adjacent blades and the cylinder wall.

6. The automatic bone cement mixing and injection device of claim 1, wherein a nut is fixed on the sliding block; a screw rod is fixed on an output shaft of the piston driving device; the nut is installed on the screw rod.
